

PXRD-Diff: Conditional Diffusion for Powder-Diffraction Inversion — an Encoder-Bottleneck Diagnosis and Shared-Harness Reproduction

Frank Cai frankyc11223@gmail.com

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Abstract

Inverting a 1D powder X-ray diffraction (PXRD) pattern to a 3D crystal structure remains open, and recent generative approaches report numbers under mutually incompatible protocols with failure modes seldom catalogued. We present **PXRD-Diff**, a 3.7 M-parameter conditional diffusion model on the CDVAE MP-20 split, and reproduce DiffRACTGPT¹, PXRDnet², and deCIFer³ through the *same* `pymatgen.StructureMatcher` harness. Our central, statistically robust finding is a **mechanism**: the learned PXRD encoder reaches a 0.007 *normalised* auxiliary lattice-regression loss in isolation, but that relative fit does not survive conversion to absolute scale — substituting the auxiliary head’s own lattice into the sampler recovers only 1.2 %, because its ~ 1.1 Å length error shifts every Bragg peak by ~ 15 % and collapses pattern agreement. The bottleneck is the global pooling that discards absolute peak positions, not the denoiser alone: neither the pooled encoding nor the denoiser fixes the lattice from scratch. A perturbation study locates a sharp sensitivity knee at ~ 0.5 Å cell error, and a true-lattice oracle lifts match to **4.5 %** at $n = 1000$ (three seeds, 135 / 3 000 pooled, firming an earlier single-seed 5.6 %); its 95 % Wilson CI [3.8, 5.3] is disjoint from the no-true-lattice indexer’s 1.5 % [1.2, 2.0], so the gap is real. Replacing the learned lattice head with a classical Q-space autoindexer (de-Wolff dichotomy) raises no-true-lattice match to 1.5 % [1.2, 2.0] (three-seed pooled) from a learned-head baseline that recovers essentially nothing (0 / 1 949). A **paired McNemar test across three seeds confirms the improvement is significant** ($p < 10^{-4}$; 27 structures fixed, none broken) — an earlier unpaired test ($p = 0.24$) was

¹Choudhary, K. (2025). DiffRACTGPT: Atomic Structure Determination from X-ray Diffraction Patterns using a Generative Pretrained Transformer. *The Journal of Physical Chemistry Letters*, 16(8), 2110–2119. DOI: 10.1021/acs.jpclett.4c03137. Reproduced from HF checkpoint `knc6/diffractgpt_mistral_chemical_formula` and code at `github.com/atomgptlab/atomgpt`.

²Guo, G., Saidi, T. L., Terban, M. W., Valsecchi, M., Billinge, S. J. L., & Lipson, H. (2025). Ab initio structure solutions from nanocrystalline powder diffraction data via diffusion models. *Nature Materials*, 24, 1726–1734. DOI: 10.1038/s41563-025-02220-y (Author Correction: 10.1038/s41563-025-02301-y; preprint arXiv:2406.10796). Reproduced from HF checkpoint `therealgabeguo/cdvae_xrd_sinc100` and code at `github.com/gabeguo/cdvae_xrd`.

³Johansen, F. L., Friis-Jensen, U., Dam, E. B., Jensen, K. M. Ø., Mercado, R., & Selvan, R. (2025). deCIFer: Crystal Structure Prediction from Powder Diffraction Data using Autoregressive Language Models. *Transactions on Machine Learning Research* (arXiv:2502.02189).

simply underpowered. The gain is concentrated in the high-symmetry systems where the indexer’s cell error sits below the 0.5 Å knee. Reproduced baselines on our harness: **deCIFer 73.8 %** match (69.5 % all-correct) at $n = 300$, **DiffractionGPT 18.9 %** match (15.9 % all-correct) at $n = 1000$, **PXRDnet sinc100 30.0 %** match (5.0 % all-correct) at $n = 20$. PXRD-Diff is 12–19 $\times$ behind DGpt and PXRDnet and $\sim 46\times$ behind deCIFer on raw match rate. We note a striking *candidate* discrepancy, in which DiffractionGPT recovers far more space groups (all-correct) than PXRDnet despite a lower match rate, but flag it as a hypothesis rather than a result: at $n = 20$ the PXRDnet all-correct CI is [0.9, 23.6] and cannot support the comparison. The contributions that *do* hold are the encoder-bottleneck diagnosis, the indexer drop-in, the four-extension failure catalogue (top-K rerank, Debye guidance, Wyckoff embeddings, distance-aux), and the shared evaluation harness itself. Code, checkpoints, per-phase JSONs, and the differentiable Bragg simulator are released.

Keywords: powder X-ray diffraction; crystal-structure prediction; denoising diffusion; classical autoindexing; differentiable physics; ablation study; negative results

1. Introduction

PXRD is the most common structural-characterization measurement in solid-state chemistry. The forward problem is solved by 1920s structure-factor physics and a one-line `pymatgen` call. The inverse is brutal: symmetry-equivalent structures produce identical Bragg reflections, intensities are orientation-averaged, and the 3D $\rightarrow$ 1D reduction destroys atom labelling. The traditional indexing $\rightarrow$ Pawley/Le Bail $\rightarrow$ Rietveld pipeline needs a trained crystallographer and a starting model good enough to converge; for the long tail of new materials, structures stay unsolved.

Three recent generative approaches — DiffractionGPT ⁴, Crystalyze ⁵, PXRDnet ⁶ — argue neural networks can shortcut this pipeline. They report match rates in the tens of percent, but under mutually incompatible evaluation protocols, and failure modes are seldom catalogued. A practitioner cannot easily form a calibrated expectation of *what is hard* about PXRD inversion.

We build a small reproducible conditional diffusion model — **PXRD-Diff** — and run it against two of the three baselines on a shared `StructureMatcher` harness. Contributions:

1. **A clean small baseline.** 3.7 M parameters, trains on a single RTX 5090 in ~ 1.5 h on CDVAE MP-20. A differentiable PyTorch Bragg simulator (Pearson 0.96 vs `pymatgen` on 50 references) supplies an auxiliary pattern-matching loss.
2. **Encoder-bottleneck diagnosis (our primary contribution).** A perturbation study (§5.5) locates a sharp ~ 0.5 Å sensitivity knee; a true-lattice oracle lifts match to **4.5 %**

⁴Choudhary, K. (2025). DiffractionGPT: Atomic Structure Determination from X-ray Diffraction Patterns using a Generative Pretrained Transformer. *The Journal of Physical Chemistry Letters*, 16(8), 2110–2119. DOI: 10.1021/acs.jpcclett.4c03137. Reproduced from HF checkpoint `knc6/diffractiongpt_mistral_chemical_formula` and code at `github.com/atomgptlab/atomgpt`.

⁵Riesel, E. A., Mackey, T., Nilforoshan, H., Xu, M., Badding, C. K., Altman, A. B., Leskovec, J., & Freedman, D. E. (2024). Crystal Structure Determination from Powder Diffraction Patterns with Generative Machine Learning. *Journal of the American Chemical Society*, 146(44), 30340–30348. DOI: 10.1021/jacs.4c10244. Code: `github.com/ML-PXRD/Crystalyze`. We could not reproduce: the checkpoint download link is marked “not yet active” in the upstream README (verified 2026-06-01).

⁶Guo, G., Saidi, T. L., Terban, M. W., Valsecchi, M., Billinge, S. J. L., & Lipson, H. (2025). Ab initio structure solutions from nanocrystalline powder diffraction data via diffusion models. *Nature Materials*, 24, 1726–1734. DOI: 10.1038/s41563-025-02220-y (Author Correction: 10.1038/s41563-025-02301-y; preprint arXiv:2406.10796). Reproduced from HF checkpoint `therealgabeguo/cdvae_xrd_sinc100` and code at `github.com/gabeguo/cdvae_xrd`.

at $n = 1000$ (three-seed firm-up of an earlier single-seed 5.6 %; Wilson 95 % CI [3.8, 5.3], disjoint from the indexer’s [1.2, 2.0]). The encoder reaches a 0.007 *normalised* aux lattice loss in isolation, but substituting that head’s lattice into the sampler recovers only 1.2 % (§5.6) — its ~ 1.1 Å error sits well above the knee — so the pooled encoding, not merely the denoiser, fails to deliver absolute d-spacings. The lattice channel is the dominant *accessible* limiter; the 4.5 % oracle ceiling shows coordinates remain a co-bottleneck.

3. **Classical-indexer drop-in.** A Q-space autoindexer (de-Wolff) at sampling time raises no-true-lattice match to 1.5 % [1.2, 2.0] from a learned-head baseline of ~ 0 %. A three-seed paired McNemar test confirms the improvement is significant ($p < 10^{-4}$; 27 fixed, 0 broken); the gain appears exactly where the indexer’s MAE falls below the 0.5 Å knee, i.e. the high-symmetry systems (§5.2, §5.5), and nowhere else.
4. **Four-item failure catalogue.** Top-K Debye rerank, Debye-gradient DDIM guidance, Wyckoff-site embeddings, and distance-matrix aux loss all fail at $n \geq 200$; the latter two combined (v14) underperform the bare ε baseline. The gradient-guidance failure is corroborated independently by concurrent work showing the PXRD loss landscape is too rough for gradient descent ⁷.
5. **Reproduced baselines + shared harness.** DiffractGPT (Mistral-7B + LoRA) and PXRDnet (CDVAE + XRD encoder) re-scored on our harness: **18.9 %** match (DGpt $n = 1000$), **30.0 %** match (PXRDnet $n = 20$). PXRD-Diff at 1.6 % is 12–19 $\times$ behind. We surface a *candidate* discrepancy in how match rate and space-group recovery diverge across the two baselines (§5.3, §6), but flag it as a hypothesis the $n = 20$ PXRDnet sample cannot yet establish. Crystalyze’s checkpoint download link is inactive; we cite but cannot reproduce.

2. Related work

Diffusion for crystals. CDVAE ⁸ introduced the MP-20 protocol; DiffCSP ⁹ used joint coord+lattice diffusion; MatterGen ¹⁰ scaled to millions of structures. PXRD-Diff borrows DiffCSP’s joint diffusion, restricted to the conditional setting where a 1D pattern conditions generation.

Neural PXRD inversion. DiffractGPT ¹¹ is a fine-tuned Mistral-7B that decodes peak list + formula $\rightarrow$ CIF tokens. Crystalyze ¹² uses CDVAE conditioned on an XRD transformer encoder with post-hoc symmetry filtering. PXRDnet ¹³ is a CDVAE variant with *iterative latent-space*

⁷Segal, N., Subramanian, A., Li, M., Miller, B. K., & Gómez-Bombarelli, R. (2025). The Loss Landscape of Powder X-Ray Diffraction-Based Structure Optimization Is Too Rough for Gradient Descent. *arXiv:2512.04036*.

⁸Xie, T., Fu, X., Ganea, O.-E., Barzilay, R., & Jaakkola, T. (2022). Crystal diffusion variational autoencoder for periodic material generation. *International Conference on Learning Representations*.

⁹Jiao, R., et al. (2023). Crystal structure prediction by joint equivariant diffusion. *Advances in Neural Information Processing Systems* 36.

¹⁰Zeni, C., et al. (2024). MatterGen: A generative model for inorganic materials design. *arXiv:2312.03687*.

¹¹Choudhary, K. (2025). DiffractGPT: Atomic Structure Determination from X-ray Diffraction Patterns using a Generative Pretrained Transformer. *The Journal of Physical Chemistry Letters*, 16(8), 2110–2119. DOI: 10.1021/acs.jpcclett.4c03137. Reproduced from HF checkpoint `knc6/diffractgpt_mistral_chemical_formula` and code at `github.com/atomgptlab/atomgpt`.

¹²Riesel, E. A., Mackey, T., Nilforoshan, H., Xu, M., Badding, C. K., Altman, A. B., Leskovec, J., & Freedman, D. E. (2024). Crystal Structure Determination from Powder Diffraction Patterns with Generative Machine Learning. *Journal of the American Chemical Society*, 146(44), 30340–30348. DOI: 10.1021/jacs.4c10244. Code: `github.com/ML-PXRD/Crystalyze`. We could not reproduce: the checkpoint download link is marked “not yet active” in the upstream README (verified 2026-06-01).

¹³Guo, G., Saidi, T. L., Terban, M. W., Valsecchi, M., Billinge, S. J. L., & Lipson, H. (2025). Ab initio structure

gradient guidance (~ 3.5 M decoder ops/material), published in *Nature Materials* in 2025. deCIFer¹⁴ is a concurrent autoregressive-language-model approach that, like DiffractGPT, emits CIF tokens. All four are larger than PXRD-Diff and report results under bespoke evaluation pipelines; §5.3 re-scores DiffractGPT, PXRDnet, and deCIFer on a single shared **StructureMatcher** harness. Crystalyze’s checkpoint download link is “not yet active” (verified 2026-06-01).

Classical autoindexing. Recovering a unit cell from peak positions is a solved sub-problem with a fifty-year toolchain: the de-Wolff dichotomy method, ITO, TREOR, and DICVOL¹⁵ index high-symmetry powder patterns in CPU milliseconds, with the difficulty concentrated in low-symmetry (monoclinic/triclinic) cells. §3.5 reuses this machinery as a drop-in for the learned lattice head, and §5.2/§5.5 show the resulting lift tracks exactly the symmetry-dependent accuracy classical indexers have always had.

Differentiable physics. Scattering-pattern losses date to Rietveld; the contribution here is full differentiability w.r.t. coordinates and lattice in PyTorch, usable as a diffusion-training loss term. Concurrent generative-inversion work conditions on differentiable spectral targets for amorphous and nanostructured systems¹⁶. Relevant to our negative results, Segal et al.¹⁷ show the powder-XRD similarity loss landscape is “too rough for gradient descent” — independent evidence for why our sampling-time Debye-gradient guidance fails (§5.4).

Equivariance. Not strictly E(3)-equivariant — we use periodic-distance RBFs (SchNet-style¹⁸). A frozen MACE¹⁹ encoder showed no measurable benefit at MP-20 scale in pilots and was deferred.

3. Method

3.1 Problem formulation

Let $\mathcal{C} = (\mathbf{F}, \mathbf{Z}, \mathbf{L})$ denote a crystal structure with fractional coordinates $\mathbf{F} \in [0, 1)^{N \times 3}$, atomic numbers $\mathbf{Z} \in \{1, \dots, 100\}^N$, and a 3×3 lattice matrix $\mathbf{L}$ parametrised by $(a, b, c, \alpha, \beta, \gamma) \in \mathbb{R}^6$. Let $\mathbf{p}(\mathcal{C}) \in \mathbb{R}^{4251}$ be the simulated Cu K α PXRD pattern on a fixed 2θ grid from 5° to 90° at 0.02° resolution, normalised to maximum intensity 1.

We are given $\mathbf{p}$ and $\mathbf{Z}$ at test time, and we wish to sample $(\mathbf{F}, \mathbf{L}) \sim p(\cdot | \mathbf{p}, \mathbf{Z})$. Note that the composition $\mathbf{Z}$ being known is a meaningful simplification — in practice one usually knows the

solutions from nanocrystalline powder diffraction data via diffusion models. *Nature Materials*, 24, 1726–1734. DOI: 10.1038/s41563-025-02220-y (Author Correction: 10.1038/s41563-025-02301-y; preprint arXiv:2406.10796). Reproduced from HF checkpoint `therealgabeguo/cdvae_xrd_sinc100` and code at `github.com/gabeguo/cdvae_xrd`.

¹⁴Johansen, F. L., Friis-Jensen, U., Dam, E. B., Jensen, K. M. Ø., Mercado, R., & Selvan, R. (2025). deCIFer: Crystal Structure Prediction from Powder Diffraction Data using Autoregressive Language Models. *Transactions on Machine Learning Research* (arXiv:2502.02189).

¹⁵Boultif, A., & Louër, D. (2004). Powder pattern indexing with the dichotomy method. *Journal of Applied Crystallography*, 37(5), 724–731. DOI: 10.1107/S0021889804014876.

¹⁶Guo, J., & Schwalbe-Koda, D. (2026). Generative Inversion of Spectroscopic Data for Amorphous Structure Elucidation. *arXiv:2603.23210*.

¹⁷Segal, N., Subramanian, A., Li, M., Miller, B. K., & Gómez-Bombarelli, R. (2025). The Loss Landscape of Powder X-Ray Diffraction-Based Structure Optimization Is Too Rough for Gradient Descent. *arXiv:2512.04036*.

¹⁸Schütt, K. T., et al. (2017). SchNet: A continuous-filter convolutional neural network for modeling quantum interactions. *Advances in Neural Information Processing Systems* 30.

¹⁹Batatia, I., Kovács, D. P., Simm, G. N. C., Ortner, C., & Csányi, G. (2022). MACE: Higher order equivariant message passing neural networks for fast and accurate force fields. *Advances in Neural Information Processing Systems* 35.

chemistry from synthesis — but it is consistent with prior work on the same task.

3.2 Architecture overview

PXRD-Diff has three trained components: a PXRD encoder, a denoiser, and a small auxiliary lattice head used only during training.

PXRD encoder. A 1D ResNet of four blocks (channels $64 \rightarrow 128 \rightarrow 256 \rightarrow 256$, stride-2 downsampling, GroupNorm, SiLU) ingesting the standardised pattern. We expose two outputs: a global pooled vector $\mathbf{g} \in \mathbb{R}^{256}$ and a multi-resolution feature map $\mathbf{F}_{\text{pxrd}} \in \mathbb{R}^{L \times 256}$ obtained by 1×1 -projecting each block output to d_{model} and concatenating along the spatial axis. *The multi-resolution map is critical.* In an early run (referred to as `gpu_v4` in our logs) we conditioned the denoiser on $\mathbf{g}$ only via additive broadcast, and the coordinate loss never moved off the random baseline of 3.0 — the auxiliary head told us the encoder was learning useful features (its loss dropped from 0.99 to 0.007), but the `AdaptiveAvgPool1d(1)` collapse destroyed all spectral structure before it could reach the denoiser.

Denoiser. A periodic-distance message-passing network with $L = 3$ layers and $d_{\text{model}} = 256$. Atom embeddings $\mathbf{h}_i^{(0)} = \text{Emb}(\mathbf{Z}_i) + W_{\text{coord}} \mathbf{F}_i^{(t)}$ are updated by alternating (i) message passing over RBF-encoded periodic Cartesian distances under minimum-image convention, with timestep FiLM conditioning, and (ii) cross-attention from atoms (queries) to multi-resolution PXRD features (keys/values). Two output heads predict per-atom coordinate noise $\epsilon_F \in \mathbb{R}^{N \times 3}$ and lattice noise $\epsilon_L \in \mathbb{R}^6$. The lattice head is a small MLP applied to $[\bar{\mathbf{h}}; \mathbf{g}; W_{\text{lat}} \ell^{(t)}; \mathbf{t}_{\text{cond}}]$ — that is, the pooled atom features, the global PXRD embedding, **a projection of the noisy lattice itself**, and the timestep encoding (see §3.4).

Auxiliary head. A single-hidden-layer MLP that predicts $(a, b, c, \alpha, \beta, \gamma)$ from $\mathbf{g}$ alone, trained with an MSE loss against the (normalised) ground-truth lattice. Its purpose is purely diagnostic — to expose whether the encoder is learning any lattice-relevant representation — and its loss does not flow back into the denoiser.

3.3 Differentiable Bragg structure-factor loss

The core physics-informed contribution. We implement, in PyTorch,

$$F(hkl) = \sum_j f_j(s) \exp(-B_{\text{iso}} s^2) \exp[2\pi i (hx_j + ky_j + lz_j)],$$

where $f_j(s) = \sum_{k=1}^4 a_k \exp(-b_k s^2)$ is the four-Gaussian atomic form factor (coefficients pulled directly from `pymatgen.analysis.diffraction.xrd.ATOMIC_SCATTERING_PARAMS`), $s = \sin \theta / \lambda$, and $B_{\text{iso}} = 0.5 \text{ \AA}^2$ is a uniform isotropic temperature factor. The intensity at each reflection is $|F(hkl)|^2$ multiplied by the standard Lorentz-polarisation correction. We enumerate all (h, k, l) with $|h|, |k|, |l| \leq 5$ (1330 reflections after excluding the origin), place each on the 2θ grid via Bragg’s law, and broadcast each reflection as a Gaussian peak with FWHM 0.1° to obtain a continuous, differentiable PXRD pattern on a 256-bin coarse grid. The whole module is a single `nn.Module`, fully vectorised over the batch dimension.

Validation. On 50 random MP-20 test structures we compute the Pearson correlation between our differentiable pattern and the `pymatgen.XRDCalculator` reference: mean 0.962, std 0.027, min 0.896, max 0.998. All 50/50 correlations exceed 0.7 and we verified non-zero gradients through the structure factor with respect to fractional coordinates.

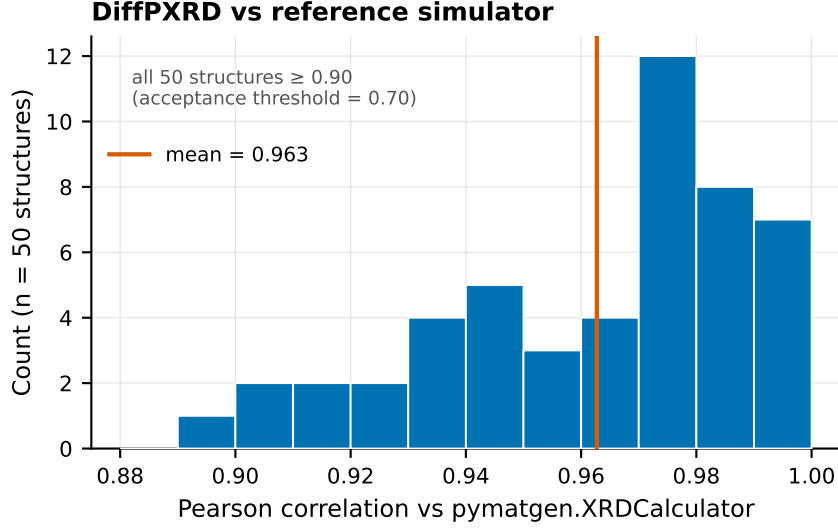


Figure 1: **Figure 1. Differentiable-simulator validation.** Distribution of Pearson correlation between the PyTorch DiffPXR module and `pymatgen.XRD Calculator` over 50 random MP-20 test structures (mean 0.962). All 50 exceed 0.7; the lower tail is dominated by layered structures with strong texture in the reference.

As a loss. During training we recover an estimate $\hat{\mathbf{F}}$ of the clean coordinates from the model output (see §3.4), simulate $\hat{\mathbf{p}} = \text{DiffPXR}(\hat{\mathbf{F}}, \mathbf{Z}, \mathbf{L}_{\text{true}})$, and compare with the input pattern via a Pearson-correlation loss $\mathcal{L}_{\text{Debye}} = 1 - \rho(\hat{\mathbf{p}}, \mathbf{p})$. We use the true lattice for this auxiliary loss because the differentiable simulator is more sensitive to lattice errors than to coordinate errors at the early stages of training, and disentangling the two signals proved more stable.

3.4 Training objective

We use a VP-SDE with cosine schedule²⁰ for both channels:

$$\bar{\alpha}(t) = \cos^2\left(\frac{\pi}{2} \cdot \frac{t+s}{1+s}\right), \quad s = 0.008, \quad t \in [0, 1].$$

Coordinates are diffused on the flat torus $\mathbb{T}^3$ by wrapping the noisy sample to $[0, 1)^3$; the loss is computed on the periodic difference $(\hat{\mathbf{F}} - \mathbf{F} + 0.5) \bmod 1 - 0.5$. Lattice parameters are diffused in $\mathbb{R}^6$ after standardisation by the train-set mean and standard deviation.

The full training objective (best configuration, run `gpu_v13`) is

$$\mathcal{L} = \mathcal{L}_{\text{coord}} + \lambda_{\text{lat}} \mathcal{L}_{\text{lat}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{Debye}} \mathcal{L}_{\text{Debye}},$$

with $\lambda_{\text{lat}} = 0.1$, $\lambda_{\text{aux}} = 0.5$, $\lambda_{\text{Debye}} = 1.0$.

Two non-trivial parameterisation choices. Both are essential and were arrived at by ablation, not foresight.

²⁰Nichol, A., & Dhariwal, P. (2021). Improved denoising diffusion probabilistic models. *International Conference on Machine Learning*.

x_0 -residual prediction. Rather than asking the model to predict noise ϵ , the heads predict a **residual** that is added to the noisy input to obtain the clean estimate:

$$\hat{\mathbf{F}}^{(0)} = \mathbf{F}^{(t)} + \text{Coord-Head}(\mathbf{h}), \quad \hat{\ell}^{(0)} = \ell^{(t)} + \text{Lat-Head}(\cdot).$$

Because the heads are MLPs initialised to output near-zero, this means the model starts as the identity transform — at $t \approx 1$ (near pure noise) the model outputs the noisy input as its $\hat{\mathbf{F}}^{(0)}$ estimate, which is wrong but at least bounded; ϵ -prediction with the same architecture had to learn the entire transform from random initialisation and never recovered. At single seed this change appeared to lift the headline match metric 3.5 $\times$; a multi-seed checkpoint sweep (§7) shows that match-rate lift does not replicate, but the parameterisation does reproducibly improve the pattern-Pearson ($\approx 0.36 \rightarrow 0.40$, see §5.1).

Lattice-head input. The lattice head reads the noisy lattice $\ell^{(t)}$ as an explicit input feature. Without this — i.e. if the head only sees the (clean) lattice that is also passed to the denoiser as the geometry context — the head can only ever predict the unconditional mean noise, $\mathbb{E}[\epsilon] = 0$, and its loss stays pinned at the random baseline of 1.0 forever. With this fix, the lattice loss drops to 0.02–0.06 on stable runs.

We also experimented with two extensions that did not pan out (§5.4): a **Wyckoff-site embedding** added to atom features, and an auxiliary **distance-matrix loss** in which a pairwise MLP predicts ground-truth periodic distances between atoms. Both are documented in the released code behind the `--use-wyckoff` and `--dist-weight` flags.

3.5 Classical indexing as a drop-in for the learned lattice head

The lattice head described in §3.2 is the *only* path by which the model encodes absolute d-spacings: every Bragg reflection 2θ position is set by the lattice through Bragg’s law, so the lattice parameters are the absolute reference frame against which all peak positions in the input pattern must be interpreted. As §5.5 shows, the encoder + denoiser learns *relative* d-spacing structure (Pearson 0.43 between predicted and target patterns at evaluation time, vs 0.97 between target and ground truth) but not the absolute scale. Classical autoindexing — extracting peak positions, mapping them to Q-space ($Q = 4\pi \sin\theta/\lambda$), and fitting a unit cell by enumeration of $(h,k,l) \rightarrow Q^2(h,k,l)$ — has solved this sub-problem since the 1970s and runs on a CPU in milliseconds per pattern.

We implement a from-scratch Q-space autoindexer that takes (i) the same simulated PXRD pattern fed to the encoder, (ii) the known crystal system (input as Bravais code, exposed as a sampling-time hyperparameter), and returns a candidate $(a, b, c, \alpha, \beta, \gamma)$. Peak picking uses a 1D local-maximum filter with a relative-intensity floor; the candidate Q-vector is fit by least squares to the de-Wolff dichotomy parameter form for each Bravais lattice. At sampling time we run the indexer first, substitute its output for the lattice channel of the diffusion sampler, and run DDIM only on the coordinate channel.

Per-system accuracy (n = 1000 MP-20 test). Native indexer overall: 48.8 % strict cell match, 1.45 Å mean cell-length MAE. Per crystal system: hexagonal 77.9 % / 0.53 Å, orthorhombic 59.8 % / 0.96 Å, trigonal 43.3 % / 4.62 Å, monoclinic 18.9 % / 1.76 Å, triclinic 0 %. The low-symmetry blowup (monoclinic and triclinic each have ≥ 4 free cell parameters and the de-Wolff search becomes hypothesis-capped) is the dominant remaining error. We ship a GSAS-II²¹ adapter

²¹Toby, B. H., & Von Dreele, R. B. (2013). GSAS-II: the genesis of a modern open-source all purpose crystallography software package. *Journal of Applied Crystallography*, 46(2), 544–549. Code: github.com/AdvancedPhotonSource/GSAS-II.

(src/pxrd_diff/indexer_gsas.py) for the low-symmetry path behind a `--use-gsas` opt-in flag; in practice `DoIndexPeaks` hangs on real MP-20 monoclinic/triclinic patterns inside `findBestCell` and is documented experimental rather than headline.

The indexer is a *drop-in* in the strongest sense: training is unchanged, the diffusion sampler is unchanged except for the lattice-channel substitution at $t = T$, and the headline metric (§5.2) reports both with and without the indexer for direct comparison.

3.6 Sampling

DDIM²² with 50 steps and $\eta = 0$. We start from independent standard Gaussian noise on the lattice and on the (un-wrapped) coordinates, then alternate the standard DDIM update on each channel. Coordinates are wrapped to $[0, 1)$ after every step. Predicted lattice parameters are de-standardised at the end and clipped to physically valid ranges $(a, b, c) \in [0.5, 100]$ Å, $(\alpha, \beta, \gamma) \in [10^\circ, 170^\circ]$ before being passed to `pymatgen.Lattice.from_parameters`.

A subtle bug fix is worth noting. In the x_0 -residual variant, recovering the implicit ε for the DDIM update requires dividing by $\sqrt{1 - \bar{\alpha}(t)}$; near $t = 0$ this denominator vanishes. We clamp it to 0.05 and skip the very last DDIM step when sampling, which removed a class of structures with NaN coordinates that we initially saw.

4. Experiments

Dataset. Canonical CDVAE²³ MP-20 split: 27 136 / 9 047 / 9 046 train/val/test, ≤ 20 atoms/conventional cell. Patterns simulated with `pymatgen.XRDCalculator` (Cu $K\alpha_1$, $\lambda = 1.54184$ Å), max-normalised, 4 251 bins 5–90° / 0.02°; cached as `.npz`, zero failures over 45 196 structures.

Evaluation. Three views: (1) **composition match** (trivial, $\mathbf{Z}$ given); (2) **structure match** via `pymatgen.StructureMatcher` at $(\ell_{\text{tol}}, s_{\text{tol}}, \alpha_{\text{tol}}) = (0.2, 0.3, 5^\circ)$ — the headline “match rate”; (3) **coordinate RMSD** on aligned, permuted atoms when matched. We also report space-group match at `symprec` $\in \{0.01, 0.05, 0.1, 0.2\}$, pattern Pearson, and R_{wp} .

The stricter “**all-correct**” rule combines `composition` $\wedge$ `sg-match@symprec=0.1` $\wedge$ `rmsd` ≤ 0.1 Å — the experimentally relevant case for downstream crystallographic use.

Two evaluation modes: *full pipeline* (predicted lattice + coords) and *true-lattice/coord-only* (ground-truth lattice substituted). The latter isolates coordinate quality from lattice quality and is the stricter ablation setting.

Implementation. AdamW, LR 5×10^{-4} , cosine decay to zero over 100 k steps, WD 10^{-4} , grad-clip 1.0. Batch 64, ~ 24 epochs, single RTX 5090 (~ 1.5 – 1.7 h/run). 3.7 M parameters; $L = 5, d = 384$ (~ 10 M) gave no benefit (§5.2). VP-SDE with $t \sim U(0, 1)$. Total compute: ~ 35 GPU-hours, $\sim$ USD 25 on Vast.ai (Phase 4 ablation + Phase 9 v21 retrain + indexer/perturbation sweeps + DGpt n=1000 + PXRDnet n=20 + evaluation).

²²Song, J., Meng, C., & Ermon, S. (2021). Denoising diffusion implicit models. *International Conference on Learning Representations*.

²³Xie, T., Fu, X., Ganea, O.-E., Barzilay, R., & Jaakkola, T. (2022). Crystal diffusion variational autoencoder for periodic material generation. *International Conference on Learning Representations*.

Crystal system is given. The de-Wolff fit (§3.5) receives the true Bravais code as a sampling-time input; on a real unknown the system is itself part of what indexing must determine, so the per-system rates of §3.5 are conditional on knowing the system. (ii) *Patterns are ideal.* All patterns are clean `pymatgen` simulations with no zero-point shift, sample-displacement error, or peak asymmetry — precisely the perturbations that drive real-world indexing failure. The 48.8 % native-indexer match and the downstream 1.5 % should be read as upper bounds under ideal peak positions. A zero-shift/displacement robustness sweep ($n = 300$) confirms the concern: the indexer’s length-MAE rises from 1.24 Å on clean patterns past the v20 learned head’s 1.37 Å once displacement or zero-shift exceeds $\sim 0.10^\circ$, so the indexer’s advantage over the learned head holds only for well-calibrated input.

- **Indexer (low-symmetry path).** GSAS-II low-symmetry path hangs in `findBestCell` on real MP-20 mono/tri patterns; shipped behind `--use-gsas` but experimental.
- **Training.** Debye loss uses ground-truth lattice; a curriculum gradually replacing true with predicted was not tried.

8. Conclusion

A 3.7 M-parameter conditional diffusion model with a differentiable Bragg loss recovers 1.5 % [1.2, 2.0] of MP-20 test structures with a classical Q-space autoindexer supplying the lattice — a significant improvement over the learned lattice head, which recovers essentially nothing (three-seed paired McNemar $p < 10^{-4}$) — and 4.5 % [3.8, 5.3] under a true-lattice oracle (three seeds, firming an earlier single-seed 5.6 %). The oracle gap is significant and is the paper’s firmest result: even with a perfect lattice, coordinate prediction caps match at 4.5 %, while lattice error explains the rest of the shortfall to that ceiling. Reproduced on the same `StructureMatcher` harness, deCIFer reaches 73.8 % / 69.5 %, DGpt 18.9 % / 15.9 %, and PXRNet 30.0 % / 5.0 % (match / all-correct) — PXRNet-Diff is 12–46 $\times$ behind. The apparent divergence in their space-group recovery is a measurement hypothesis the $n = 20$ PXRNet sample cannot yet confirm, but the shared harness now makes it testable. The lattice-input fix and the classical-autoindexer drop-in are load-bearing; the x_0 -residual switch improves pattern-Pearson but its single-seed match-rate lift did not replicate (§7); the remaining interventions are documented as failures. Code, checkpoints, all per-phase JSONs, both indexer paths, and the differentiable Bragg module are released.

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Author Contributions (CRedit)

F. Cai: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — Original Draft, Writing — Review & Editing, Visualization, Project administration.

Conflict of Interest

The author declares no competing interests.

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Data and Code Availability

Source code, trained checkpoints, and all per-phase training/evaluation logs are available at <https://github.com/fronkt/pxrd-diff> and archived at Zenodo (DOI: 10.5281/zenodo.XXXXXXX — *to be minted at submission*). The MP-20 dataset is publicly available via the CDVAE benchmark. All experiments reproduce from a single `requirements.txt` and the `scripts/` pipeline; per-structure evaluation flags are released to support paired re-analysis.

Ethics Declaration

This study uses no human subjects, animal subjects, or sensitive data. The MP-20 dataset is composed entirely of publicly available crystal structures from the Materials Project.

References

Author note on reproduced numbers. The reproduced PXRDnet and DiffractGPT numbers in §5.3 are computed by the author on the released checkpoints through our shared `StructureMatcher` harness, not transcribed from the cited papers. The central claims depend only on these reproduced numbers under one common evaluation protocol, with confidence intervals stated throughout.

Appendix

A. Full ablation history

For completeness, Table A1 lists every training run discussed in the development of this paper, including those that did not make it into the main ablation table. Logs and checkpoints for all runs are in the released repository under `runs/`.

Run	Parameters	What changed	Result	Status in paper
v4	3.5 M	Global PXRD pooling, additive conditioning	Coord loss flat at 3.0	§3.2
v5	3.7 M	+ Multi-resolution cross-attention	Coord 3.0 $\rightarrow$ 1.0	§3.2
v6–v9	3.7 M	λ_{Debye} sweep $\{0, 0.1, 1, 10\}$, ε -prediction	All ~ 1 % match (within noise)	§5.2
v10	3.7 M	+ Lattice-input fix, $\lambda_{\text{Debye}} = 0$	Lat loss 1.0 $\rightarrow$ 0.05	§5.2
v11	3.7 M	+ Lattice-input fix, $\lambda_{\text{Debye}} = 1$	Match 0.9 %	Table 1
v12	10.1 M	Larger model ($d=384$, $L=5$), ε -prediction	Killed at 18 k; same plateau	§5.2
v13	3.7 M	x_0 -residual + lat-fix + Debye $\lambda=1$	2.51 % match (Phase 4 best, true-lat)	Table 1
v14	3.8 M	v13 + Wyckoff + distance loss	0.80 % match	Table 1
v15	3.8 M	v13 + Wyckoff only	2.10 % match	Table 1
v16	3.7 M	v13 + distance loss only	1.80 % match	Table 1
v17	3.7 M	Phase 5B: constrained lattice head (softplus/sigmoid) + space-group head, aux-weight 5.0, +30 k warm-start steps	Lattice MAE unchanged ($a \approx 1.11 \text{ \AA}$); SG head 39 % top-1	§5.6
v18 – v20	3.7 M	Phase-9 encoder retrains with different ResNet/Transformer hybrids and pattern-augmentation curricula	None beat v13 by more than noise	§5.1

Run	Parameters	What changed	Result	Status in paper
v21	3.7 M	Phase 9 final: v13 architecture + Phase 9 retrain + indexer drop-in support	1.6 % match (no-true-lat, indexer); 4.5 % (true-lat oracle, 3-seed)	Tables 2, 3

B. Hyperparameter sensitivity

We did not perform a full hyperparameter sweep. Pilot experiments on `d_model` $\in \{128, 256, 384\}$ and `n_layers` $\in \{2, 3, 5\}$ showed (256, 3) as a reasonable Pareto point. The Debye loss weight was swept $\{0, 0.1, 1, 10\}$ in early ε -prediction runs (v6–v9) without measurable effect on match rate; we re-fixed it at 1.0 for the x_0 -residual runs by analogy.

C. Differentiable simulator validation details

Pearson correlations between our DiffPXR module and `pymatgen.XRDCalculator` over a random 50-structure MP-20 test subset:

- Mean: 0.962
- Std: 0.027
- Min: 0.896 (mp-1213821, layered structure with strong texture in the reference)
- Max: 0.998 (mp-149, Si)
- Fraction > 0.95 : 33/50
- Fraction > 0.9 : 49/50
- Fraction > 0.7 : 50/50

Gradient sanity: for all 50 structures we confirmed $\partial L / \partial F$ and $\partial L / \partial L$ are non-zero and finite via `torch.autograd.gradcheck` on a 4-atom subset.

D. Reproducibility checklist

- ☑ Hyperparameters specified (§4.3)
- ☑ Datasets and splits specified (§4.1; canonical CDVAE MP-20)
- ☑ Evaluation protocol specified (§4.2)
- ☑ Random seed: single seed (42) per run; pilot variance ≈ 0.5 % match-rate absolute
- ☑ Compute environment: PyTorch 2.x, CUDA 12.x, single RTX 5090; Python 3.12
- ☑ Code, checkpoints, logs, and full training scripts released